ALTEN

AI TECH LEAD & ARCHITECT

Core Mission: Bridging the gap between business ambition and production-grade AI systems.

STRATEGIC FRAMING & VISION (CLARIFY & IDENTIFY)

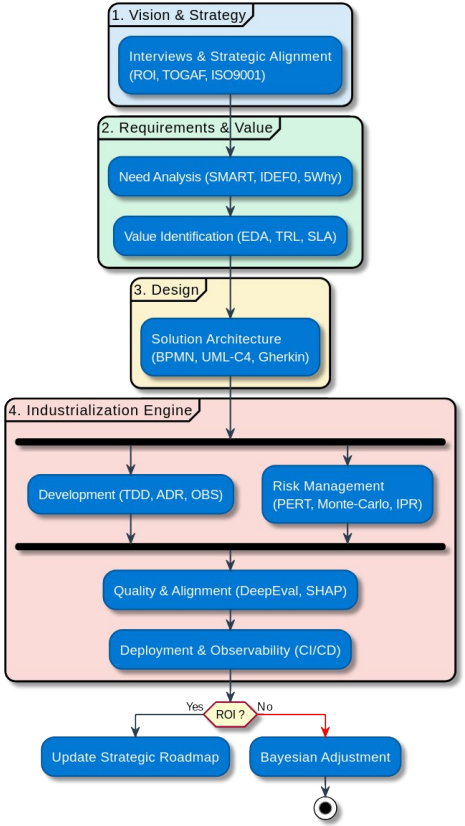
Expert in Industry & Services, I translate complex operational needs into technical reality within highly constrained environments (isolated networks, security mandates). My approach focuses on identifying value and securing the end-to-end lifecycle, deploying industrial-grade solutions such as multi-agent systems and semantic ontologies.

ENGINEERING EXCELLENCE & STANDARDIZATION (BUILD & DESIGN)

I drive AI industrialization through robust MLOps/LLMOps and scalable architectural design (DAT). Leveraging deep R&D and Industrial expertise, I bridge the gap between research and production, focusing on standardization, real-time execution, and Edge/Hybrid computing. I ensure every solution is built for reliability, security, and seamless integration into complex hardware and software ecosystems.

TECHNICAL LEADERSHIP & DELIVERY (ROADMAP & DEV)

I provide technical and operational steering for AI projects, ensuring a seamless transition from scoping to production. I manage the complete product lifecycle—from V-Model (Safran, Automotive, etc.) to Agile (Airbus, Thales, etc.)—ensuring technical consistency and mission success for major industrial accounts.



TECH STACK
(CLOUD & EDGE AGNOSTIC)

CORE AI: LangGraph, LlamaIndex, Qdrant, vLLM, Ollama, Transformer (ViT), PyTorch, Sklearn.
DATA & ETL: SQLAlchemy, Polars, DuckDB, LanceDB, ArangoDB, Mage-ai, PydanticAI, Unstructured.io.
ROBOTICS & EDGE: Dora-rs, CAN, MQTT.
SECURITY & IAM: LiteLLM, HashiCorp Vault, Keycloak.
OPS & EVAL: SHAP, Arize Phoenix, DeepEval, MLflow, Evidently.
UI & API: Chainlit, FastAPI, React, Marimo.
CI/CD & QUALITY: GitLab CI/CD, Pydantic, SonarQube.
INFRA: Kubernetes (K8s) & Helm, Docker, OpenStack.
SAFETY: Architecture "HIL" & Quantification (MAPIE).

Domain	Core Competencies & Experience
Tech Skills	**Python**: 12y (SOLID, TDD) • **Data**: 10y (ACID, MLOps) • **Vision**: 5y (DL) • **GenAI**: 5y (RAG, Agents) • **DevOps**: 5y (K8s, CI)
Sectors	**Science & Research**: 5y (PhD) • **Industry & Defense**: 5y
Methodology	**Leadership**: 4y • **AI Architecture**: 3y (IAF/TOGAF) • **Lifecycle**: 4y (Agile/V-Model)

FROM IMAGERY TO ACTIONABLE INTELLIGENCE

Accelerating satellite data exploitation for faster military decisions (AI Architect - Defense)

CLIENT CONTEXT & PROBLEM

High-frequency satellite imagery (IMINT) saturation exceeds manual processing limits. This data-to-intelligence gap creates a bottleneck, resulting in critical decision latency in near-real-time (NRT) cycles.

WHY PRIORITIZED

Strategic decision loop compression. Accelerating the transition from raw sensor data to structured intelligence is a prerequisite for maintaining tactical initiative in high-intensity environments.

SOLUTION & RATIONALE

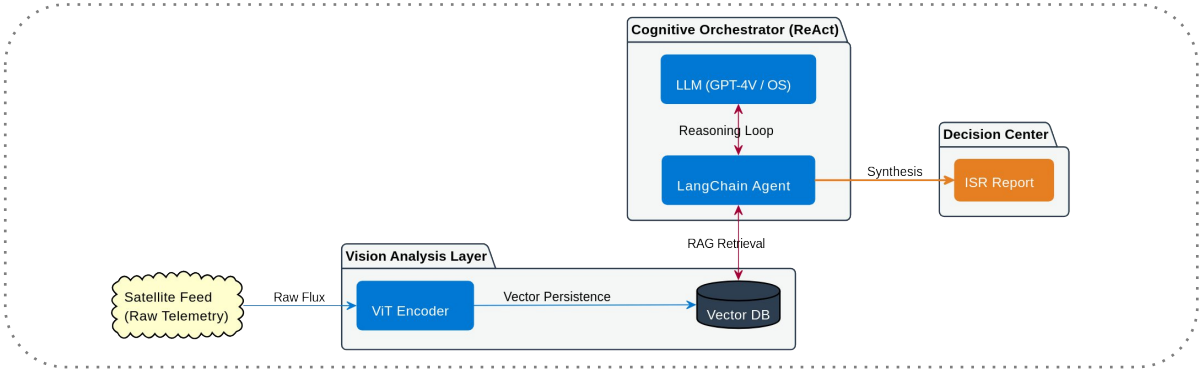
Asynchronous pipeline using ViT encoders for semantic feature extraction. ReAct agentic loops cross-reference detections with tactical knowledge bases via vector search to synthesize reports.

TECHNOLOGIES

Multimodal LLMs (GPT-4V / Apache 2.0 OS), LangChain, Python, Gitlab CI/CD, Vector DB.

Architecture

Asynchronous pipeline utilizing ViT encoders for semantic feature extraction. Persisted vector embeddings enable contextual retrieval, while ReAct agentic loops cross-reference detections with tactical ontologies to synthesize deterministic reports.

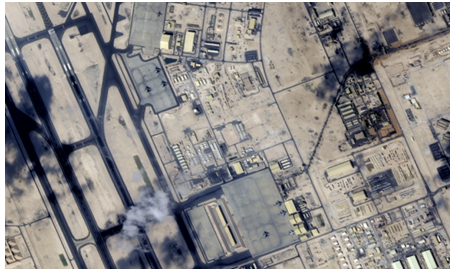


Value / KPI

Demonstrated reduction in report latency and a increase in operational throughput per analyst.

Investment

2-month scoping and project launch phase. Focused on automated report generation and architectural alignment.



OPTIMIZING MISSION LOGISTICS DECISIONS

AI-driven allocation under complex operational constraints (AI Tech Lead - Defense)

CLIENT CONTEXT & PROBLEM

High-intensity fleet mobility exceeds manual logistical planning capacity. Resource allocation (spare parts, maintenance) under unstable constraints results in sub-optimal mission readiness.

WHY PRIORITIZED

Strategic Paris Air Show showcase. Demonstrating operational superiority through "right stocks, right place, right time" optimization and a reduced logistical footprint while maximizing fleet availability.

SOLUTION & RATIONALE

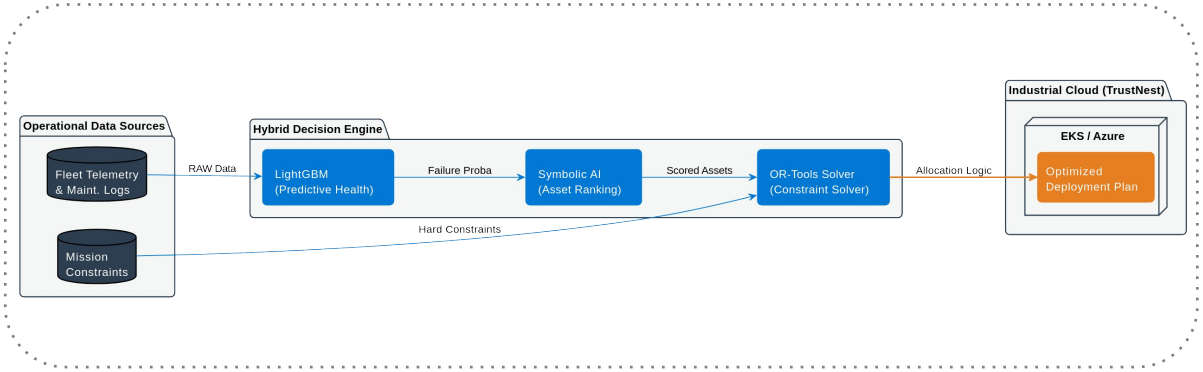
Operations Research (OR) + ML framework. OR-Tools manage deterministic constraints and rules-driven allocation, while ML/Deep Learning models handle predictive maintenance and failure detection.

TECHNOLOGIES

OR-Tools, LightGBM, Symbolic AI (Expert), Azure, EKS (Kubernetes), Gitlab CI.

Architecture

Ingestion of operational data processed by LightGBM for predictive failure modeling. OR-Tools solve for optimal asset ranking and spare part inventory right-sizing based on mission-specific parameters.



Value / KPI

Reduction in logistical costs and automated calculation of mission success probabilities.

Investment

6 months | Industrialization focused on high-resilience and "TrustNest Restricted" deployment.

SPATIAL LOCALIZATION OF BLADE DEFECTS

3D-to-2D registration for automated engine inspection (AI Tech Lead & Architect - Aerospace)

CLIENT CONTEXT & PROBLEM

Boroscopic images exhibit high perspective distortion, preventing accurate mapping of 2D detections to 3D CAD models. Manual localization is insufficient for precise structural integrity assessment.

WHY PRIORITIZED

Required for high-fidelity damage tracking. Determining exact 3D coordinates is mandatory to monitor defect evolution and validate repair tolerances in turbine blade lifecycles.

SOLUTION & RATIONALE

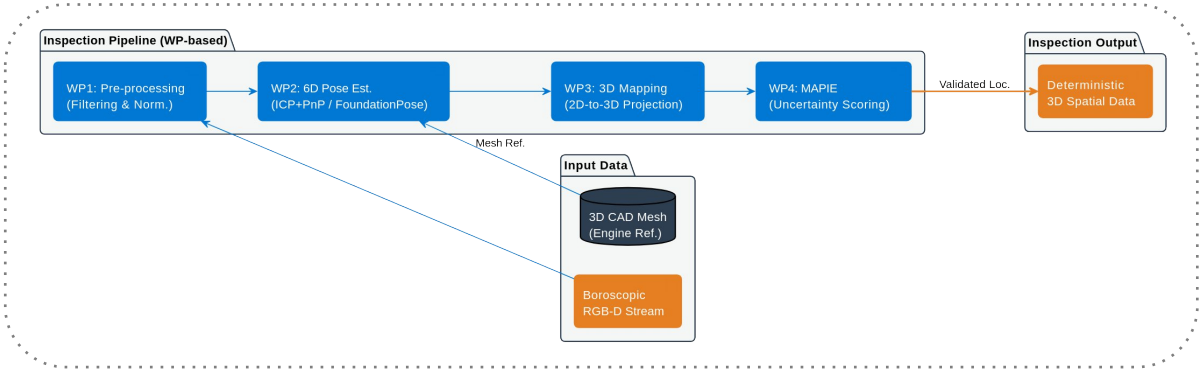
3D-to-2D registration PoC using boroscopic RGB-D data. Evaluation of Geometric (ICP + PnP) vs. Neural (FoundationPose) pipelines. Integration of MAPIE to provide rigorous uncertainty quantification for the estimated 6D pose.

TECHNOLOGIES

FoundationPose, ICP, PnP, MAPIE, PyTorch3D, Trimesh, Open3D.

Architecture

Modular 4-Work Package (WP) structure. The pipeline estimates the camera pose relative to the mesh and projects 2D pixels into 3D space, followed by a statistical validation layer.



Value / KPI

Millimeter-level localization accuracy with quantified confidence intervals. Transition from qualitative checks to deterministic 3D spatial data.

Investment

4-month PoC | 4 Work Packages.

SOVEREIGN EDGE AI FOR VEHICLE DIAGNOSTICS

Secure LLM deployment for confidential automotive data (AI Project Leader - Automotive)

CLIENT CONTEXT & PROBLEM

Automated interpretation of multi-protocol vehicle telemetry (CAN / OBD-II) under strict confidentiality. Cloud-based analysis is prohibited due to industrial IP risks and data privacy mandates.

WHY PRIORITIZED

Absolute Digital Sovereignty. Secure, air-gapped diagnostic capabilities are required for sensitive industrial missions where zero cloud access is permitted.

SOLUTION & RATIONALE

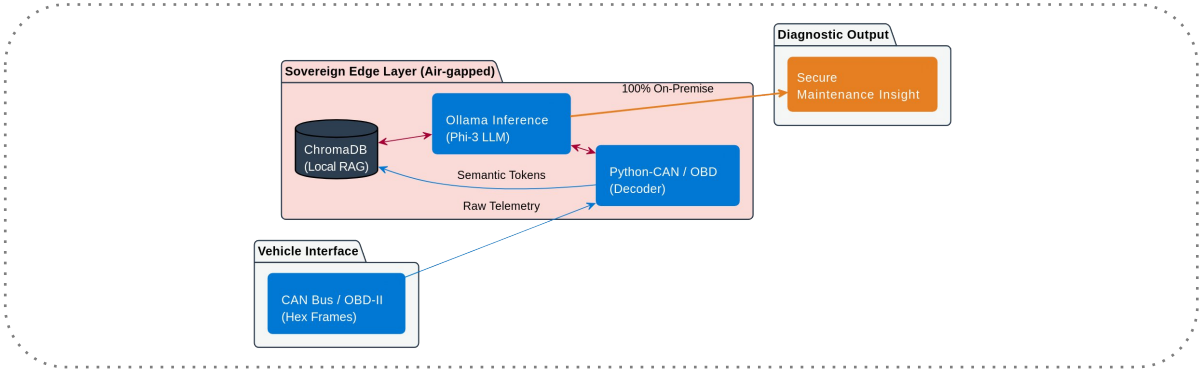
Local LLM deployment using Ollama/Phi-3. A hybrid decoding pipeline translates raw hexadecimal frames and OBD-II PID codes into semantic tokens for privacy-preserving reasoning.

TECHNOLOGIES

Ollama, Phi-3, ChromaDB, Python-CAN, Python-OBD, FastAPI, Docker.

Architecture

Real-time ingestion of CAN/OBD frames via a Python-CAN and OBD parser. Data is indexed in ChromaDB for local RAG, allowing the model to perform diagnostic inference without external connectivity.



Value / KPI

Real-time vehicle diagnostics with 100% on-premise execution, securing industrial opportunities in highly restricted environments.

Investment

5 months | Edge AI.

SCALING INDUSTRIAL IMPACT ANALYSIS WITH GRAPHS

Managing complex dependencies at production scale (AI Architect – Aerospace)

CLIENT CONTEXT & PROBLEM

Change impact analysis across multiple Bills of Materials (xBOM) is hindered by rigid relational database structures. Navigating deep recursive dependencies (10M+ nodes) causes significant performance bottlenecks.

WHY PRIORITIZED

Risk of production line blockage. Rapidly calculating the ripple effect of a technical change is critical to maintaining industrial flow and preventing costly configuration errors.

SOLUTION & RATIONALE

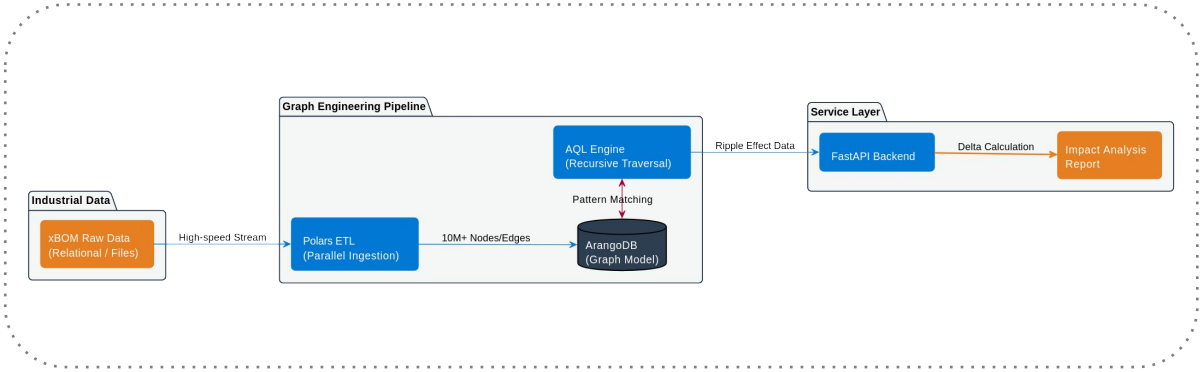
Migration to a Graph Database (ArangoDB) model. This natively handles complex industrial ontologies and many-to-many relationships, enabling real-time traversal of massive dependency trees.

TECHNOLOGIES

ArangoDB, Polars, FastAPI, Docker, AQL.

Architecture

High-speed ingestion of xBOM data via Polars for parallelized ETL. Data is modeled into ArangoDB, where AQL (ArangoDB Query Language) performs recursive impact analysis, served via a FastAPI backend.

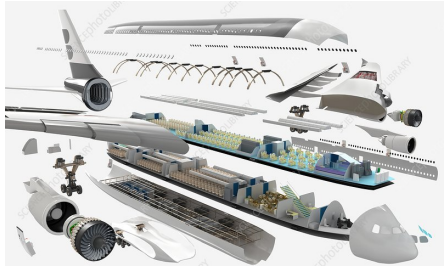


Value / KPI

Full automation of configuration delta calculations and reduction in manual cross-referencing errors.

Investment

5-month development cycle | High-scale Data Engineering.



INDUSTRIALIZING AI AT SCALE

Building reliable and auditable GenAI systems (AI Tech Lead & Architect – AI Factory)

CLIENT CONTEXT & PROBLEM

Transitioning from stochastic PoCs to production-grade services often accumulates technical debt. Scaling requires addressing non-deterministic model behaviors and the lack of type-safety in agentic workflows.

WHY PRIORITIZED

Requirement for auditable AI lifecycles to ensure secure and resilient GenAI deployments under enterprise-grade loads.

SOLUTION & RATIONALE

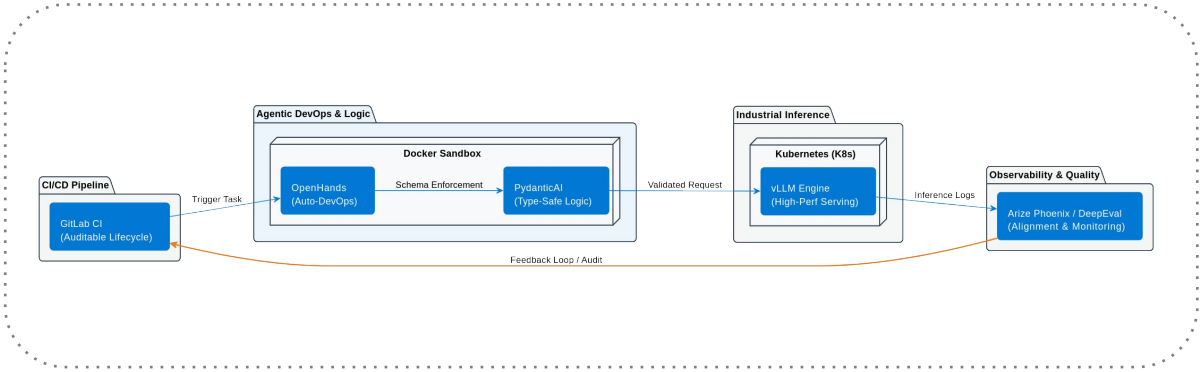
Engineering framework based on SOLID principles and TDD. Implementation of Agentic DevOps: using PydanticAI for schema-enforced, type-safe reasoning and OpenHands within Docker sandboxes to automate development and deployment tasks.

TECHNOLOGIES

PydanticAI, OpenHands, Kubernetes, vLLM, Docker, Arize Phoenix, DeepEval, GitLab CI.

Architecture

Containerized stack where OpenHands manages autonomous tasks. Logic built with PydanticAI for runtime validation of model outputs. Inference via vLLM on Kubernetes, monitored by Arize Phoenix and DeepEval for RAG quality and alignment.

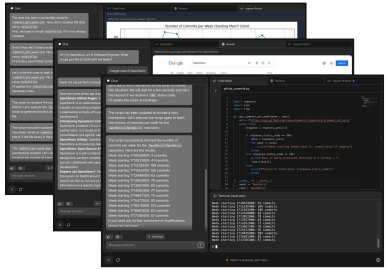


Value / KPI

Reduction in Time-to-Market (goal). Long-term operational sustainability through standardized architectures and technical mentorship.

Investment

6-month Architect mandate. (in progress)



ACCELERATED DYNAMICS FOR ROBOTIC AUTONOMY

Real-time VLA orchestration for autonomous vehicle control (AI Architect – AI Factory)

CLIENT CONTEXT & PROBLEM

Integrating Vision-Language-Action (VLA) policies into robotic hardware often introduces prohibitive latency. Traditional middleware overhead prevents the fluid execution of natural language instructions in dynamic environments.

WHY PRIORITIZED

High-stakes R&D for autonomous control. The objective is to achieve low-latency, end-to-end robotic autonomy where high-level linguistic commands are translated into precise physical actions in real-time.

SOLUTION & RATIONALE

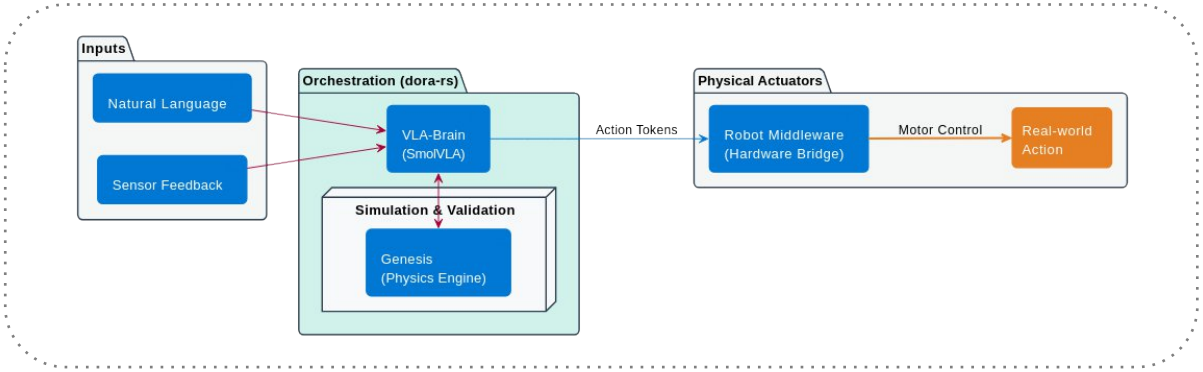
Orchestration via dora-rs for its high-performance, zero-copy communication capabilities. This is paired with a SmolVLA policy (LeRobot) to maintain a small memory footprint while ensuring rapid inference on edge hardware.

TECHNOLOGIES

dora-rs, Genesis Physics Engine, PyTorch, LeRobot (SmolVLA), Docker, uv.

Architecture

Hierarchical control loop where the VLA-Brain processes multi-modal inputs. The system interfaces with the Genesis Physics Engine for high-fidelity simulation, utilizing dora-rs for efficient message passing between the control terminal and the physical actuators.



Value / KPI

Achieved a stable closed-loop control pipeline at 20Hz (50ms latency).

Investment

Agile R&D Project | Full-stack Robotics & AI. (In progress)